

Dipak Shanki

Kathmandu, Nepal

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Summary

AI/ML fresher with a strong interest in developing intelligent AI solutions, focused on Agentic AI, AI automation workflows, Generative AI, and multimodal systems.

Core Skills

- **Languages:** Python, JavaScript.
- **Libraries & Frameworks:** PyTorch, Hugging Face Transformers, NumPy, Pandas
- **AI/ML:** Deep Learning Fundamentals (CNNs, Transformers, Sequence Modeling), Generative AI Basics, Text-to-Speech (TTS), Lip-Sync Video Generation, Model Training & Evaluation, Data Preparation & Evaluation Metrics.
- **Engineering:** End-to-End AI Pipeline Development, Audio & Video Processing.
- **Tools:** Git, Jupyter Notebook

AI Research & Projects

1. End-to-End Multimodal Lip-Sync Pipeline

Python, PyTorch, Wav2Lip, SpeechT5, OpenCV

- Built a text/audio-to-video pipeline by integrating **SpeechT5** (speech synthesis) with **Wav2Lip** (lip-sync generation).
- Developed modular preprocessing and inference for face detection, mel-spectrogram generation, and audio-visual alignment.
- Optimized inference reliability and memory usage through batch-size and resolution tuning.

GitHub: github.com/dipak-shaaki/LipSync_Project

2. TTS Benchmarking Pipeline

Python, PyTorch, Hugging Face Transformers

- Built a reproducible benchmarking workflow for **SpeechT5** to evaluate synthesis latency and perceived audio quality.
- Implemented automated audio generation and logging across test prompts, with structured result export to CSV for analysis.
- Added inference-stability improvements (fixed speaker embedding seed and end-silence padding) to reduce output variance and audio cutoff.
- **Outcome:** Created a reusable baseline framework for selecting TTS models in low-latency content generation pipelines.

GitHub: github.com/dipak-shaaki/TTS_Benchmark

Work Experience

AI/ML Engineer

2026 March – 2026 Aug

PalmMind Technology, Lalitpur, Nepal

- Developed and improved AI-powered chatbot solutions for multiple insurance companies using Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs).
- Implemented REST APIs with FastAPI and integrated embedding models, vector databases, and similarity search for accurate information retrieval.
- Enhanced chatbot capabilities by creating guided conversation flows, FAQ systems, and product knowledge pipelines using structured CSV datasets.
- Optimized inference performance, prompt engineering, and response quality while reducing latency and improving retrieval accuracy.
- Collaborated using Git in an Agile development environment and participated in testing, debugging, and deployment of production-ready AI services.

Certifications

1. Commonwealth Bank – Data Science Job Simulation (Forage)

2025
- Conducted large-scale data aggregation and cleaning of customer datasets, ensuring data privacy through anonymization techniques.
 - Designed relational database schemas to optimize query performance for structured financial data analysis.
2. AI/ML Bootcamp

2024
- Hands-on training on supervised learning, data preprocessing, model evaluation, and real-world ML use cases using Python.

Education

Nagarjuna College of IT	Kathmandu, Nepal
Bachelor of Science in Computer Science and Information Technology (BSc CSIT)	2022 – 2026
Sudurpaschimanchal Academy	Dhangadhi, Nepal
Higher Secondary Education (+2 Science)	Completed 2021